ATROPINE SULFATE

[ATROPEN, SAL-TROPINE]

Anticholinergic | Muscarinic antagonist | Systemic antispasmodic



1 MECHANISM OF ACTION (MOA) & TARGET

- **TARGET:** Muscarinic acetylcholine receptors
- **MOA:** Competitively blocks acetylcholine
  - ➡ ⬇ parasympathetic tone
  - ➡ ⬆ heart rate & ⬇ secretions
- **Analogy:** Cutting the brake lines on the vagus nerve to accelerate cardiac output and dry respiratory secretions.



2 FORMS & STRENGTHS

- **Inj:** 0.1 mg/mL, 1 mg/10 mL
- **Auto-Injector:** 2 mg
- **Ophth Sol:** 1%



3 INDICATIONS (FDA-APPROVED & MAJOR OFF-LABEL)

- **FDA-APPROVED:**
  - Symptomatic bradycardia
  - Organophosphate/carbamate poisoning
  - Pre-anesthesia antisialagogue
- **MAJOR OFF-LABEL:**
  - PEA/asystole (historical ACLS)
  - Cholinergic crisis • Mydriasis for eye exams
  - Pylorospasm • Mushroom poisoning

4 DOSAGE & ADMINISTRATION

A ADULT DOSE

- Bradycardia: 1 mg IV q3-5min
- Max: 3 mg total
- **Route:** IV/IO/IM • Push rapidly
- Avoid doses <0.5 mg (paradoxical bradycardia)

B PEDIATRIC DOSE

- Bradycardia: 0.02 mg/kg IV q3-5min
- Min: 0.1 mg • Max: 0.5 mg per dose
- **Route:** IV/IO/IM • Push rapidly

C SPECIAL ADJUSTMENT

- **Renal:** No adjustment needed
- **Hepatic:** Use caution
- **Organophosphate poisoning:** 2–6 mg IV, double q5min until atropinization (secretions dry, HR ↑)

D ADMINISTRATION

- **Route:** IV/IO/IM
- **Instructions:** Push rapidly for bradycardia
- Repeat every 3–5 min as needed

5 PHARMACOKINETICS & DYNAMICS (PK/PD)

- **Onset:** 1 min (IV)
- **T1/2:** 2–3 hrs
- **Metabolism:** Hepatic hydrolysis to tropine & tropic acid

6 ADVERSE EFFECTS & TOXICITY

- **Common:** Dry mouth, blurred vision, tachycardia, urinary retention, photophobia, constipation
- **Severe:** Hyperthermia, delirium, arrhythmias

7 CONTRAINDICATIONS & INTERACTIONS

- **ABSOLUTE CONTRAINDICATIONS:**
  - Narrow-angle glaucoma
  - Obstructive uropathy
  - Severe ulcerative colitis

BOXED WARNING: NONE

- **MAJOR INTERACTION:** Anticholinergics ➡ additive toxicity ➡ severe hyperthermia/delirium

8 SPECIAL POPULATIONS (PREGNANCY & LACTATION)

- **Pregnancy:** Use if indicated
- **Lactation:** May inhibit lactation
- **Geriatric:** Increased risk of delirium

9 CLINICAL PEARLS

- Avoid doses <0.5 mg in adults
- Antidote for organophosphates: titrate until secretions dry (atropinization)
- Crosses blood-brain barrier, causes CNS effects
- Does not treat Mobitz II or 3rd-degree AV block
- Monitor core temperature to prevent hyperthermia

10 GLOSSARY OF ABBREVIATIONS

- **ACLS** = Advanced Cardiovascular Life Support
- **AV** = Atrioventricular
- **IO** = Intraosseous
- **PEA** = Pulseless Electrical Activity
- **q3-5min** = Every 3 to 5 minutes
- **q5min** = Every 5 minutes